Run report: Orchestral Input Pipeline: Structured Prompt Conditioning for Full-Ensemble AI Music Generation

Author: KEYUR SHAILESHKUMAR PATEL (ORCID https://orcid.org/0009-0008-5261-043X)

Affiliations: (1) MSc Artificial Intelligence student, Department of Computer Science, Brunel University London, UK; (2) Independent researcher

Conducted independently; not an official Brunel University London project.

Run: 20261030-223515_full_7a8b2b82 (full mode)

Started 2026-10-30T22:35:15+00:00, finished 2026-10-31T06:20:59+00:00

Config SHA-256: 7a8b2b82226ba1fe0b6ebe6120b84108140d8ad4df0a166e24f1cc3137a80a1a

Requests SHA-256: 78b651e6688c6520c115aa3448807a3faaf9803ee6078ecc7780dc60f1f6266c

Pieces evaluated: 1152 (failed: 0)

Human participants: none (fully computational study)

Every file in this folder is listed with its SHA-256 hash in SHA256SUMS.

Results in plain words

midi: the pipeline condition scored higher than plain on prompt_adherence (median difference 0.008, 95% CI 0.005 to 0.014, Holm-adjusted p = 7.07e-11, effect size r = 0.957, 48 requests): a statistically significant difference.

midi: the pipeline condition scored lower than plain on section_clarity (median difference -0.167, 95% CI -0.222 to -0.067, Holm-adjusted p = 0.559, effect size r = -0.293, 48 requests): no significant difference.

midi: the pipeline condition scored higher than plain on dynamics_accuracy (median difference 0.920, 95% CI 0.874 to 0.946, Holm-adjusted p = 1.36e-12, effect size r = 1.000, 44 requests): a statistically significant difference.

midi: the pipeline condition scored higher than plain on structure_f (median difference 0.200, 95% CI 0.111 to 0.236, Holm-adjusted p = 0.000129, effect size r = 0.772, 48 requests): a statistically significant difference.

musicgen: the pipeline condition scored lower than plain on prompt_adherence (median difference -0.002, 95% CI -0.003 to -0.000324, Holm-adjusted p = 0.015, effect size r = -0.507, 48 requests): a statistically significant difference.

musicgen: the pipeline condition scored equal than plain on section_clarity (median difference 0.000, 95% CI -0.062 to 0.056, Holm-adjusted p = 1.000, effect size r = 0.014, 48 requests): no significant difference.

musicgen: the pipeline condition scored lower than plain on dynamics_accuracy (median difference -0.084, 95% CI -0.181 to 0.017, Holm-adjusted p = 0.720, effect size r = -0.271, 44 requests): no significant difference.

musicgen: the pipeline condition scored higher than plain on structure_f (median difference 0.097, 95% CI 0.000 to 0.111, Holm-adjusted p = 1.000, effect size r = 0.152, 48 requests): no significant difference.

stable_audio: the pipeline condition scored lower than plain on prompt_adherence (median difference -0.003, 95% CI -0.005 to -0.001, Holm-adjusted p = 0.015, effect size r = -0.510, 48 requests): a statistically significant difference.

stable_audio: the pipeline condition scored higher than plain on section_clarity (median difference 0.056, 95% CI -0.095 to 0.125, Holm-adjusted p = 1.000, effect size r = 0.129, 48 requests): no significant difference.

stable_audio: the pipeline condition scored lower than plain on dynamics_accuracy (median difference -0.009, 95% CI -0.117 to 0.132, Holm-adjusted p = 1.000, effect size r = 0.026, 44 requests): no significant difference.

stable_audio: the pipeline condition scored equal than plain on structure_f (median difference 0.000, 95% CI -0.067 to 0.083, Holm-adjusted p = 1.000, effect size r = -0.075, 48 requests): no significant difference.

Mean scores by generator and condition

generator	condition	prompt_adherence	section_clarity	dynamics_accuracy	structure_f	structure_chance	judge_overall	dynamic_range_db	range_violations
midi	pipeline	0.007	0.515	0.911	0.571	0.391	7.000	21.645	0.000
midi	plain	-0.006	0.588	0.004	0.397	0.391	7.000	1.745	96.597
musicgen	expert	0.007	0.477	-0.044	0.484	0.391	7.000	16.051	-
musicgen	pipeline	0.004	0.474	0.024	0.438	0.391	7.000	10.289	-
musicgen	plain	0.006	0.473	0.090	0.416	0.391	7.000	12.645	-
stable_audio	expert	0.005	0.469	-0.061	0.412	0.391	7.000	36.812	-
stable_audio	pipeline	0.002	0.487	-0.004	0.393	0.391	7.000	30.674	-
stable_audio	plain	0.005	0.450	0.002	0.413	0.391	7.000	27.627	-

structure_chance is the boundary score expected from random boundaries; structure_f should be read against it.

Confirmatory tests (preregistered)

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
confirmatory	midi	pipeline vs plain	prompt_adherence	48	0.007	-0.002	0.008	0.005	0.014	25.000	6.42e-12	7.07e-11	True	0.957
confirmatory	midi	pipeline vs plain	section_clarity	48	0.490	0.632	-0.167	-0.222	-0.067	398.500	0.080	0.559	False	-0.293
confirmatory	midi	pipeline vs plain	dynamics_accuracy	44	0.942	0.005	0.920	0.874	0.946	0.000	1.14e-13	1.36e-12	True	1.000
confirmatory	midi	pipeline vs plain	structure_f	48	0.667	0.422	0.200	0.111	0.236	103.000	1.29e-05	0.000129	True	0.772
confirmatory	musicgen	pipeline vs plain	prompt_adherence	48	0.005	0.007	-0.002	-0.003	-0.000324	290.000	0.002	0.015	True	-0.507
confirmatory	musicgen	pipeline vs plain	section_clarity	48	0.494	0.469	0.000	-0.062	0.056	533.000	0.935	1.000	False	0.014
confirmatory	musicgen	pipeline vs plain	dynamics_accuracy	44	0.005	0.077	-0.084	-0.181	0.017	361.000	0.120	0.720	False	-0.271
confirmatory	musicgen	pipeline vs plain	structure_f	48	0.444	0.444	0.097	0.000	0.111	347.500	0.399	1.000	False	0.152
confirmatory	stable_audio	pipeline vs plain	prompt_adherence	48	0.004	0.006	-0.003	-0.005	-0.001	288.000	0.002	0.015	True	-0.510
confirmatory	stable_audio	pipeline vs plain	section_clarity	48	0.500	0.467	0.056	-0.095	0.125	491.500	0.443	1.000	False	0.129
confirmatory	stable_audio	pipeline vs plain	dynamics_accuracy	44	-0.027	0.021	-0.009	-0.117	0.132	482.000	0.885	1.000	False	0.026
confirmatory	stable_audio	pipeline vs plain	structure_f	48	0.444	0.444	0.000	-0.067	0.083	325.000	0.688	1.000	False	-0.075

Secondary tests

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	musicgen	pipeline vs expert	prompt_adherence	48	0.005	0.007	-0.003	-0.004	-0.001	206.000	4.19e-05	0.000921	True	-0.650
secondary	musicgen	pipeline vs expert	section_clarity	48	0.494	0.478	0.022	-0.078	0.122	554.000	0.727	1.000	False	0.058
secondary	musicgen	pipeline vs expert	dynamics_accuracy	44	0.005	-0.006	0.101	-0.101	0.165	429.000	0.448	1.000	False	0.133
secondary	musicgen	pipeline vs expert	structure_f	48	0.444	0.444	0.000	-0.111	0.111	290.500	0.069	0.961	False	-0.325
secondary	musicgen	pipeline vs expert	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	pipeline vs expert	cultural_similarity	10	0.009	0.010	-0.001	-0.003	0.002	17.000	0.322	1.000	False	-0.382
secondary	musicgen	pipeline vs expert	dynamic_range_db	48	8.850	14.122	-3.691	-6.940	-2.233	184.000	1.25e-05	0.000287	True	-0.687
secondary	musicgen	expert vs plain	prompt_adherence	48	0.007	0.007	0.001	0.000449	0.001	361.000	0.019	0.307	False	0.386
secondary	musicgen	expert vs plain	section_clarity	48	0.478	0.469	-0.022	-0.083	0.056	539.500	0.795	1.000	False	-0.043
secondary	musicgen	expert vs plain	dynamics_accuracy	44	-0.006	0.077	-0.150	-0.264	-0.025	286.000	0.014	0.237	False	-0.422
secondary	musicgen	expert vs plain	structure_f	48	0.444	0.444	0.056	0.000	0.111	141.500	0.008	0.137	False	0.524
secondary	musicgen	expert vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	musicgen	expert vs plain	cultural_similarity	10	0.010	0.010	-0.000648	-0.002	0.001	23.000	0.695	1.000	False	-0.164
secondary	musicgen	expert vs plain	dynamic_range_db	48	14.122	10.285	2.243	0.385	4.665	329.000	0.007	0.137	False	0.440
secondary	stable_audio	pipeline vs expert	prompt_adherence	48	0.004	0.006	-0.003	-0.004	-0.000993	235.000	0.000177	0.004	True	-0.600
secondary	stable_audio	pipeline vs expert	section_clarity	48	0.500	0.450	0.014	-0.103	0.144	530.000	0.719	1.000	False	0.060
secondary	stable_audio	pipeline vs expert	dynamics_accuracy	44	-0.027	-0.124	0.079	-0.059	0.197	351.000	0.094	1.000	False	0.291
secondary	stable_audio	pipeline vs expert	structure_f	48	0.444	0.389	0.000	-0.111	0.000	354.500	0.816	1.000	False	-0.043
secondary	stable_audio	pipeline vs expert	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	pipeline vs expert	cultural_similarity	10	0.006	0.010	-0.003	-0.007	0.000366	8.000	0.049	0.732	False	-0.709
secondary	stable_audio	pipeline vs expert	dynamic_range_db	48	31.176	37.357	-7.801	-10.568	-4.408	241.000	0.000234	0.005	True	-0.590
secondary	stable_audio	expert vs plain	prompt_adherence	48	0.006	0.006	-6.07e-05	-0.001	0.002	515.000	0.461	1.000	False	0.124
secondary	stable_audio	expert vs plain	section_clarity	48	0.450	0.467	0.000	-0.044	0.083	458.000	0.666	1.000	False	0.075
secondary	stable_audio	expert vs plain	dynamics_accuracy	44	-0.124	0.021	-0.041	-0.113	0.027	368.000	0.141	1.000	False	-0.257
secondary	stable_audio	expert vs plain	structure_f	48	0.389	0.444	0.000	-0.089	0.083	360.500	0.884	1.000	False	-0.027

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	stable_audio	expert vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	expert vs plain	cultural_similarity	10	0.010	0.008	-0.000105	-0.002	0.003	27.000	1.000	1.000	False	0.018
secondary	stable_audio	expert vs plain	dynamic_range_db	48	37.357	27.661	9.543	5.341	13.237	140.000	7.71e-07	1.85e-05	True	0.762
secondary	midi	pipeline vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	midi	pipeline vs plain	cultural_similarity	10	0.010	0.001	0.007	0.003	0.014	2.000	0.006	0.029	True	0.927
secondary	midi	pipeline vs plain	dynamic_range_db	48	23.679	1.777	22.208	19.166	23.758	0.000	7.11e-15	4.26e-14	True	1.000
secondary	midi	pipeline vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	pipeline vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	pipeline vs plain	cultural_similarity	10	0.009	0.010	-0.001	-0.003	0.000887	16.000	0.275	0.279	False	-0.418
secondary	musicgen	pipeline vs plain	dynamic_range_db	48	8.850	10.285	-1.835	-5.479	0.635	375.000	0.028	0.085	False	-0.362
secondary	musicgen	pipeline vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	pipeline vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	pipeline vs plain	cultural_similarity	10	0.006	0.008	-0.004	-0.006	-0.000984	3.000	0.010	0.039	True	-0.891

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	stable_audio	pipeline vs plain	dynamic_range_db	48	31.176	27.661	3.517	-1.241	5.725	443.000	0.139	0.279	False	0.247
secondary	stable_audio	pipeline vs plain	judge_overall	48	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000

Agreement between measures (Spearman)

measure_a	measure_b	n	spearman_rho	p
prompt_adherence	section_clarity	1152	-0.000693	0.981
prompt_adherence	dynamics_accuracy	1056	-0.009	0.775
prompt_adherence	structure_f	1152	0.034	0.254
prompt_adherence	judge_dynamics	1152	-0.105	0.000345
section_clarity	dynamics_accuracy	1056	-0.005	0.875
section_clarity	structure_f	1152	-0.063	0.033
section_clarity	judge_dynamics	1152	-0.058	0.050
dynamics_accuracy	structure_f	1056	0.066	0.031
dynamics_accuracy	judge_dynamics	1056	-0.012	0.688
structure_f	judge_dynamics	1152	0.056	0.056

Intent-parser accuracy on the benchmark

Mean family F1 1.000, form correct 48/48, dynamic shape correct 48/48 (optimistic: rules were written with these requests).

Held-out requests: mean family F1 0.703, form correct 12/15, shape correct 3/15.

Frechet Audio Distance (CLAP embeddings, lower is closer to real recordings)

generator	condition	n	n_reference	fad_clap
midi	pipeline	144	52	0.023
midi	plain	144	52	0.098
musicgen	expert	144	52	0.043
musicgen	pipeline	144	52	0.027
musicgen	plain	144	52	0.028
stable_audio	expert	144	52	0.028
stable_audio	pipeline	144	52	0.011
stable_audio	plain	144	52	0.024

Limitations

All measures are automatic proxies for musical quality; no human listeners were involved.

The AI judge is itself a model with its own biases; it is reported as a secondary measure only.

CLAP similarities depend on the CLAP model's training data, which under-represents some styles (for example Indian classical music).

Pieces are short (default 30 s), so large-scale form is tested only in compressed form.

The intent parser was developed alongside the main benchmark; its accuracy there is optimistic. Parser accuracy on the held-out requests is the fair estimate.

Closed services (Suno) cannot be re-run exactly and are excluded from confirmatory tests.

Reproduce and verify

python run_all.py verify results/20261030-223515_full_7a8b2b82